EXPERIMENT - 14

14. Perform Perspective Transformation on the given image using python and Open CV.

Aim:

To perform perspective transformation on a given image using Python and OpenCV.

Procedure:

1. Import the OpenCV and NumPy libraries.
2. Read the input image using cv2.imread().
3. Select four points from the original image.
4. Define four corresponding points for the transformed image.
5. Calculate the perspective transformation matrix using cv2.getPerspectiveTransform().
6. Apply the transformation using cv2.warpPerspective().
7. Display the original and transformed images.

Program:

```
import cv2
import numpy as np

img = cv2.imread("image.jpg")

if img is None:
   print("Image not found")
else:
   rows, cols = img.shape[:2]

   pts1 = np.float32([[50, 50], [cols-50, 50],
                  [50, rows-50], [cols-50, rows-50]])

   pts2 = np.float32([[0, 0], [cols, 0],
                  [50, rows], [cols-50, rows]])

   matrix = cv2.getPerspectiveTransform(pts1, pts2)

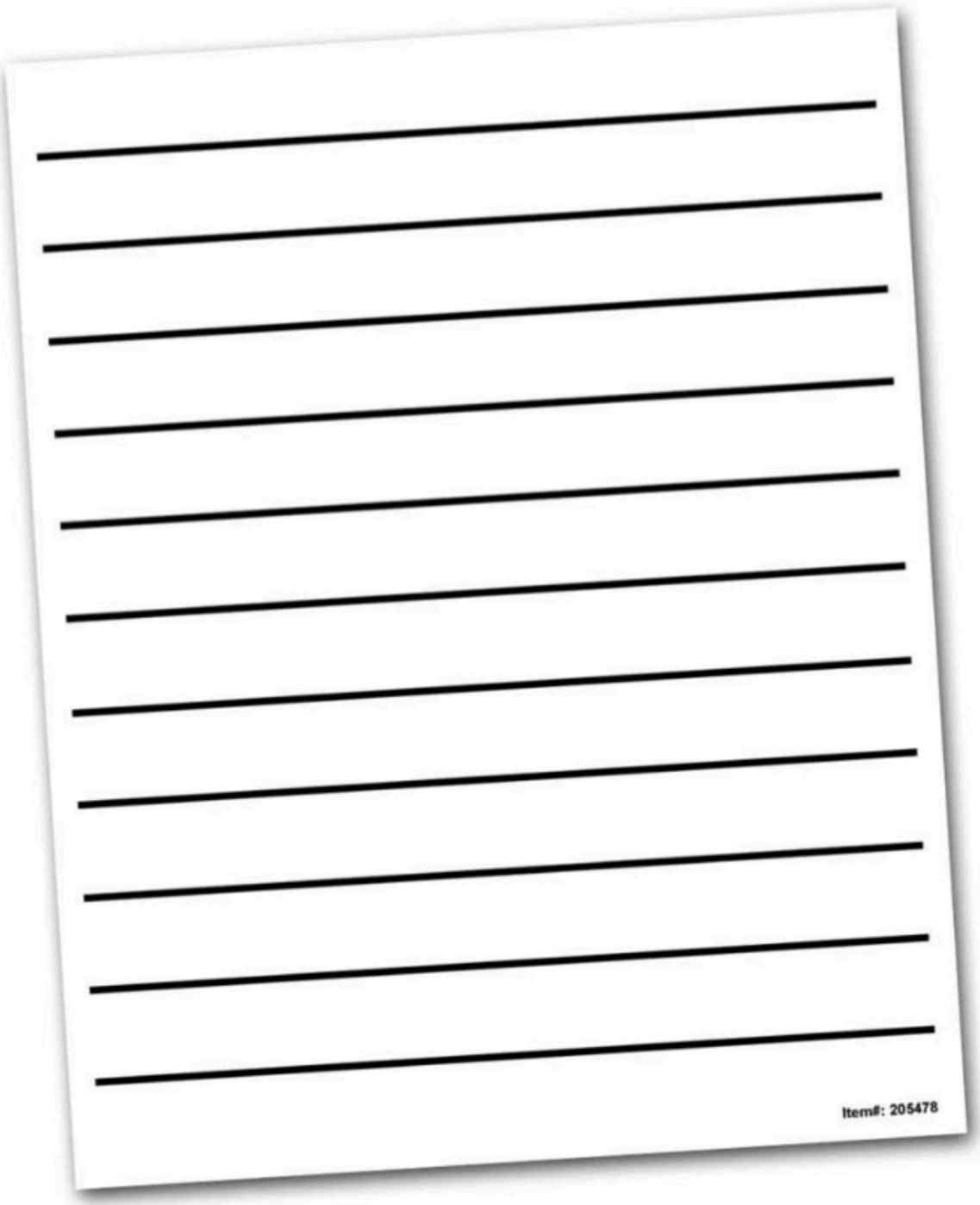
   result = cv2.warpPerspective(img, matrix, (cols, rows))

   cv2.imshow("Original Image", img)
   cv2.imshow("Perspective Transformed Image", result)

   cv2.waitKey(0)
```

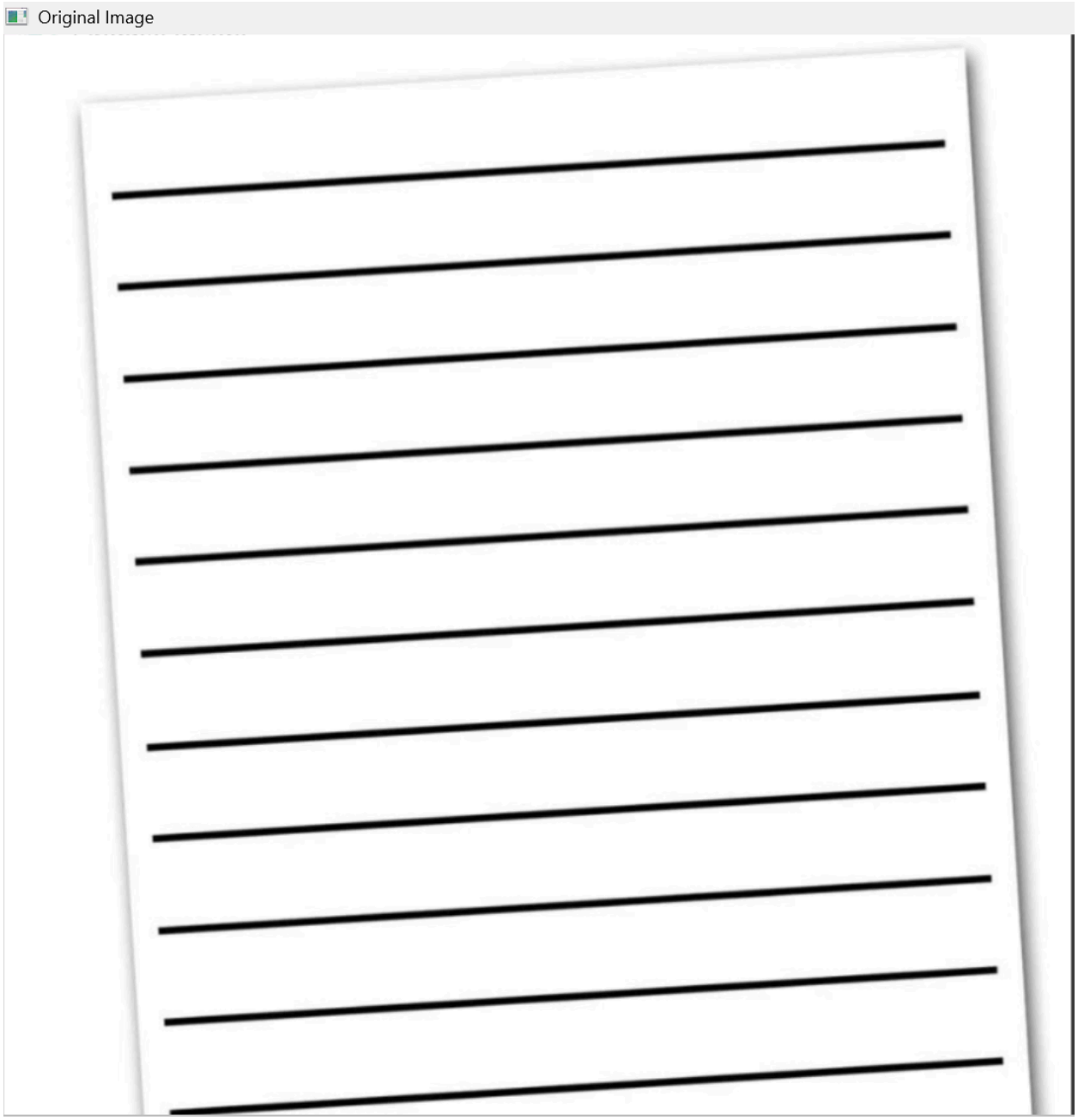
cv2.destroyAllWindows()

Sample Input:



Sample Output:

Original Image



Perspective Transformed Image



Result:

Thus, perspective transformation was successfully performed on the given image using Python and OpenCV.